



# Activity 3: Available options for study cases

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free  
litteraT



## > MOHID Lagrangian Calculation Process

1. **Interpolation** of environmental forcing variables to the tracer position.

- General: linear interpolation
- Boundary layer:  $v_{i,j} = \frac{v_{i,j}^*}{\kappa} \ln \left( \frac{x_{i,3} - H_i^b}{\varepsilon_0} \right), \quad j = 1, 2, 3$

2. Evaluation of all active **process contributions** for that tracer.

- State vector:  $\mathbf{S}_i(t) = (\mathbf{x}_i, source_i, age_i, V_i, \rho_i, \dots)$
- Process contribution:  $\mathcal{K}_i^p(\mathbf{S}_i(t), t)$

3. **Assembly** of the total kinematic tendency:  $\frac{d}{dt} \mathbf{S}_i(t) = \sum_p \mathcal{K}_i^p(\mathbf{S}_i(t), t)$

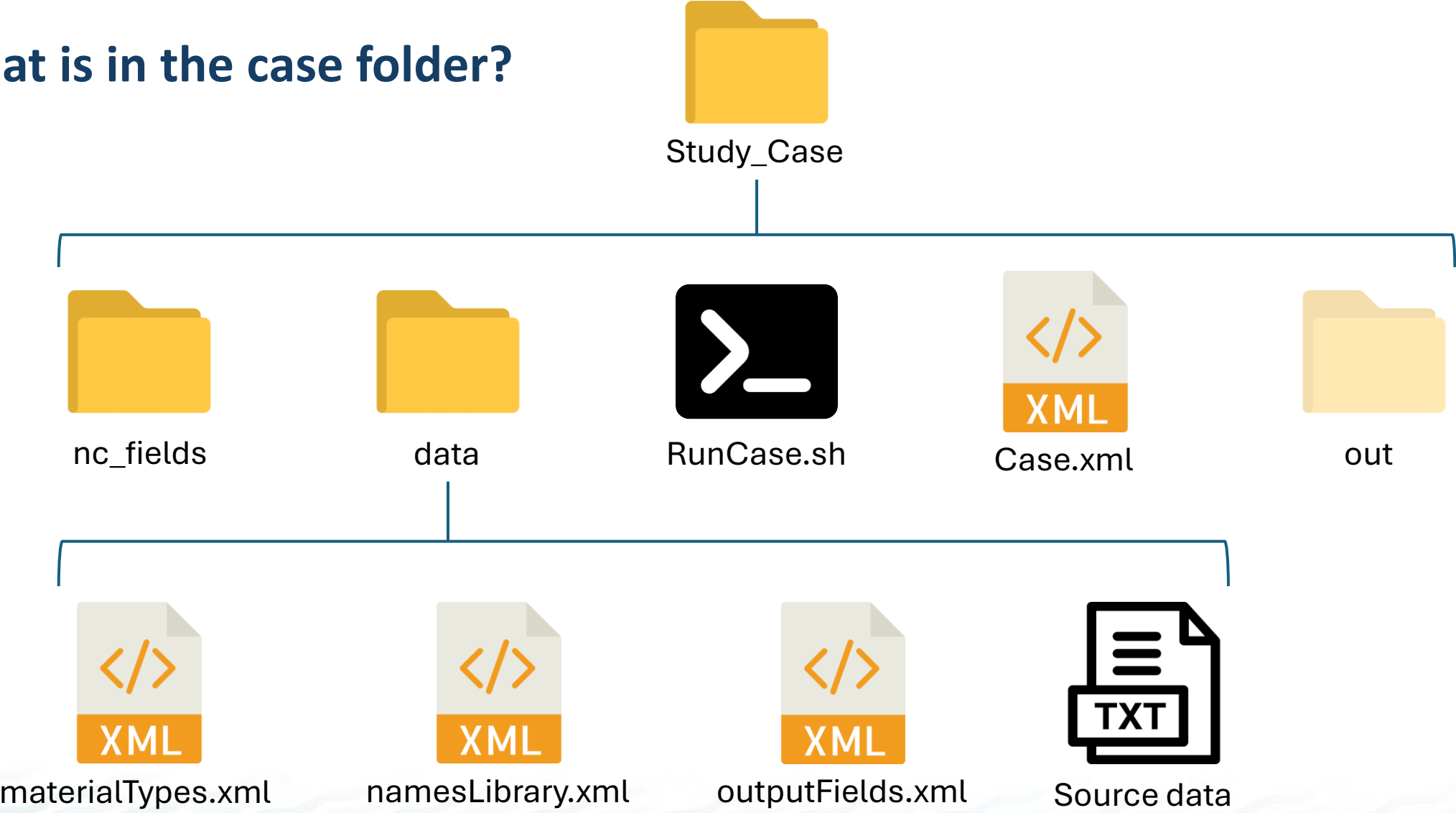
4. Time **integration**

5. **Update** of tracer state variables

Paper coming out soon!!!

**Shabani, M. ; Riveira-Varela, M. et al.**  
**(submitted)**

> What is in the case folder?





## Case.xml

```
<execution>
  <parameters>
  </parameters>

  <variableNaming>
  </variableNaming>

  <outputFields>
  </outputFields>

  <postProcessing>
  </postProcessing>
</execution>
```

```
<caseDefinitions>
  <inputData>
  </inputData>

  <simulation>
  </simulation>

  <sourceDefinitions>
  </sourceDefinitions>

  <sourceTypes>
  </sourceTypes>

  <constants>
  </constants>
</caseDefinitions>
```

## > execution → parameters

```
<parameters>
  <parameter key="Start" value="2020 01 02 00 00 00" units_comment="ISO format" />
  <parameter key="End" value="2020 01 10 00 00 00" units_comment="ISO format" />
  <parameter key="Integrator" value="1" />
  <parameter key="Threads" value="2" />
  <parameter key="BufferSize" value="86400" units_comment="seconds" />
  <parameter key="OutputWriteTime" value="3600" units_comment="seconds" />
</parameters>
```

- Integrator algorithm:
  1. Euler (Default)
  2. Multi-Step Euler
  3. RK4
- BufferSize default value: 518400s (6 days)
- Recommended: Same time lapse as input files.

## > execution → variableNaming & outputFields

```
<variableNaming>  
|   <file name="data/namesLibrary.xml"/>  
</variableNaming>  
  
<outputFields>  
|   <file name="data/outputFields.xml"/>  
</outputFields>
```



## namesLibrary.xml

### General structure

```
<naming>
  <variables>
    <variable_long_name name="name used in code">
      <variant name="alternative name 1" />
      <variant name="alternative name 1" />
    </variable_long_name>
  </variables>
  <dimensions>
    <dimension_long_name name="name used in code">
      <variant name="alternative name 1" />
      <variant name="alternative name 1" />
    </dimension_long_name>
  </dimensions>
</naming>
```

### Variable example

```
<variables>
  ...
  <eastward_sea_water_velocity name="u">
    <variant name="u" comment="used in MOHID" />
    <variant name="velocity U" comment="used in MOHID" />
    <variant name="uu" />
    <variant name="U" />
    <variant name="uo" comment="used in CMEMS" />
    <variant name="u_east" comment="used by SINTEF" />
    <variant name="velocity U" />
  </eastward_sea_water_velocity>
  ...
</variables>
```

### Dimension example

```
<dimensions>
  ...
  <longitude name="lon">
    <variant name="lon" />
    <variant name="longitude" />
  </longitude>
  ...
</dimensions>
```



## outputFields.xml

Basic output fields (always written):

- id
- source
- position (x,y,z)
- velocity (u,v,w)
- state



```
<output>
  <!-- Base tracer variables -->
  <field name="dVelX"           output="yes" />
  <field name="dVelY"           output="yes" />
  <field name="dVelZ"           output="yes" />
  <field name="VelStandardDeviation" output="yes" />
  <field name="age"             output="yes" />
  <field name="bathymetry"      output="yes" />
  <field name="rugosityVar"     output="yes" />
  <field name="D50Var"          output="yes" />
  <field name="dwz"             output="yes" />
  <field name="dist2bottom"     output="yes" />
  <!-- Other material types variables -->
  <field name="density"         output="yes" />
  <field name="radius"          output="yes" />
  <field name="volume"          output="yes" />
  <field name="area"            output="yes" />
  <field name="condition"       output="yes" />
  <field name="concentration"    output="yes" />
  <field name="initial_volume"  output="yes" />
  <field name="temp"            output="yes" />
  <field name="salt"            output="yes" />
  <field name="particulate"     output="yes" />
  <field name="radius_cr_min"   output="yes" />
  <field name="radius_cr_max"   output="yes" />
</output>
```

## > CaseDefinition → inputData

```
<inputData>
  <inputDataDir name="nc_fields/hydrodynamic" type="hydrodynamic"/>
  <inputDataDir name="nc_fields/waterProperties" type="waterProperties"/>
  <inputDataDir name="nc_fields/winds" type="meteorology"/>
  <inputDataDir name="nc_fields/waves" type="waves"/>
  <inputDataDir name="nc_fields/biogeochemistry" type="biogeochemistry"/>
</inputData>
```

- **hydrodynamic:** Hydrodynamic seawater currents (u,v,w).
- **waterProperties:** Salinity and temperature fields.
- **meteorology:** Wind fields (wind gust velocity at 10m).
- **waves:** Stokes drift velocities OR wave height, period, direction and length.
- **biogeochemistry:** Primary production, diatom and nanophytoplankton concentrations (currently under development for the biofouling module).

## > CaseDefinition → simulation

```
<simulation>
  <resolution dp="300.0" units_comment="metres (m)"/>
  <timestep dt="600.0" units_comment="seconds (s)"/>
  <BoundingBoxMin x="-9.1" y="42.39" z="-140" units_comment="(deg,deg,m)"/>
  <BoundingBoxMax x="-8.72" y="42.68" z="3" units_comment="(deg,deg,m)"/>
  <BathyminNetcdf value="1" comment="Bathymetry is a property in the netcdf. 1:true, 0:false. Default = 1"/>
  <RemoveLandTracer value="0" comment="Remove tracers on land 0:No, 1:Yes. Default = 1" />
  <VerticalVelMethod value="1" comment="1:From velocity fields, 2:Divergence based, 3:Disabled. Default = 1" />
</simulation>
```

Process activation:

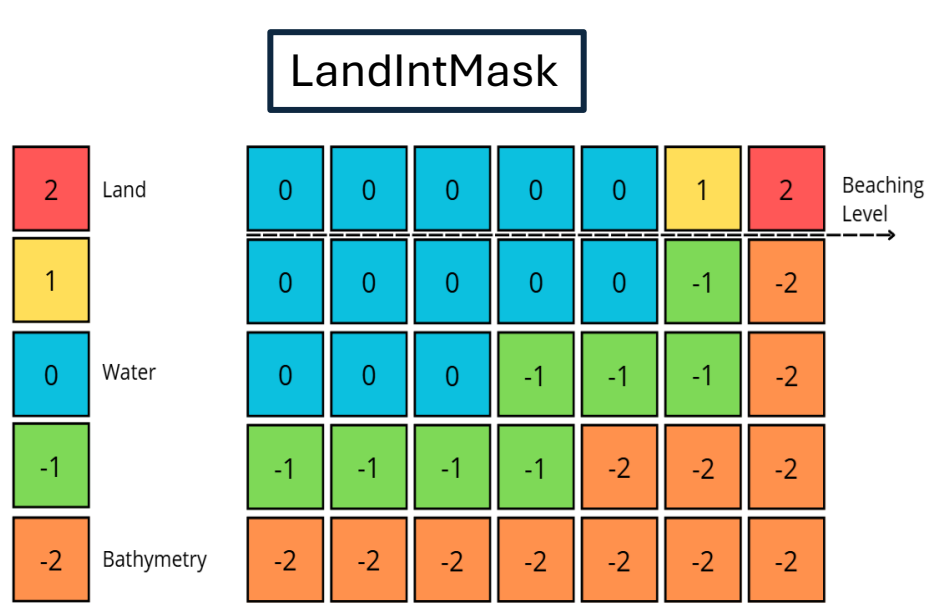
- **BathyminNetcdf:** Bathymetry is a property in the netCDF input data
- **RemoveLandTracer:** Remove tracers near/on land  $|\text{state}| > 1.9$
- **VerticalVelMethod:** Vertical velocity calculation method
- ...

## > Some basic shared processes: Diffusion

- Reynolds decomposition:  $\mathbf{v}(\mathbf{x}, t) = \bar{\mathbf{v}}(\mathbf{x}, t) + \mathbf{v}'(\mathbf{x}, t)$
- Random walk model:  $v'_{k,j} = \sqrt{3}(2r_k - 1) \cdot \alpha_j \cdot \sqrt{\frac{D}{\Delta t}}, \quad j = 1, 2, 3 \quad \left( \begin{array}{l} \alpha_{1,2} = 1 \rightarrow \text{horizontal diffusion} \\ \alpha_3 = 10^{-6} \rightarrow \text{vertical diffusion} \end{array} \right)$
- Stability criterion:  $v'_{k,j} \propto \frac{1}{\sqrt{\Delta t}} \rightarrow \min(\alpha_j \cdot \sqrt{\frac{D}{\Delta t}}, 0.1 \cdot \|v_{adv}\|)$

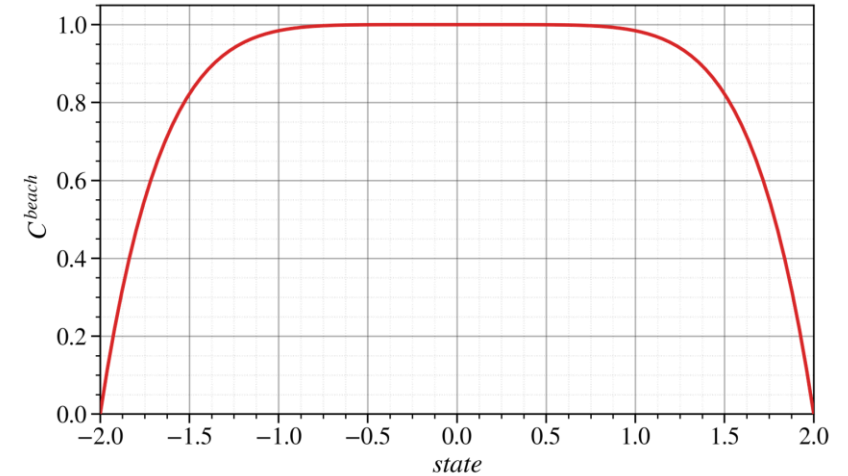
```
<DiffusionCoeff value="1.0" comment="Default = 1.0" units_comment="m2/s" />
```

## > Some basic shared processes: Boundary interaction



```
<BeachingLevel value="-3.0" units_comment="m" />
```

**Velocity reduction**

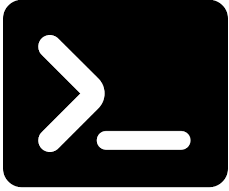


$$C_i^{beach} = 1 - \left( \frac{\text{TracerMask}_i - \text{watervalue}}{\text{landvalue} - \text{watervalue}} \right)^6$$

$$v_j^{\text{updated}}(\mathbf{x}, t) = C_i^{beach} v_j(\mathbf{x}, t), \quad j = 1, 2, 3$$

+

```
<BeachingStopProb value="80" comment="Default = 50%" units_comment="%" />
```



## RunCase.sh

- ***name*** = Base name of your definition .xml file.
- ***dirout*** = Output directory where simulation results are going to be saved.
- MOHID-Lagrangian executable and Preprocessor paths
- Copy definition .xml file and data folder into *dirout*.
- Run preprocessor (read and sort input data files time-wise)
- Run MOHID-Lagrangian executable

# Simulation time!



 ***ParaView***



# Activity 4: Source types and base case simulations

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## > General structure of an emission source

```
<sourceDefinitions>
  <source>
    <setsource id="1" name="PointRelease" />
    <rate_dt value="1" comment="number of timesteps / emission. 1 is every timestep" />
    <active start="0" end="end" units_comment="seconds (s)" />
    <point x="-8.83" y="42.60" z="0" units_comment="(deg,deg,m)" />
  </source>
</sourceDefinitions>
```

+

```
<sourceTypes>
  <types>
    <type source="1" type="base" property="" comment="passive base tracer" />
  </types>
  <file name="data/materialTypes.xml"/>
</sourceTypes>
```

## > Source types

- Point source
- Polygon from file
- Box (2D or 3D)
- Straight line
- Polyline

```
<!-- 1) point - single release location -->
<point x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
```

```
<!-- 2) polygon - release over a 2D polygon (e.g. from KMZ file) -->
<resolution x="..." y="..." z="..." units_comment="metres (m)"/>
<polygon>
  <file name="data/polygon.kmz"/>
  <verticalBoundingBox min="..." max="..." comment="z-range of release (m)"/>
</polygon>
```

```
<!-- 3) box - rectangular patch (lower-left corner + size) -->
<resolution x="..." y="..." z="..." units_comment="metres (m)"/>
<box>
  <point x="..." y="..." z="..." units_comment="(deg,deg,m)" comment="lower-left vertex"/>
  <size x="..." y="..." z="..." units_comment="metres (m)"/>
</box>
```

```
<!-- 4) line - release along a straight segment -->
<line>
  <pointa x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
  <pointb x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
</line>
```

```
<!-- 5) polyline - release along a multi-vertex path -->
<polyline>
  <point x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
  <point x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
  <point x="..." y="..." z="..." units_comment="(deg,deg,m)"/>
  ...
</polyline>
```

## > Available emission rate options

```

<!-- 1) rate_dt - emit 1 tracer every N simulation timesteps -->
<rate_dt value="1" comment="timesteps per emission. 1 = every timestep" />

<!-- 2) rate_seconds + rate_trcPerEmission - N tracers every X seconds -->
<rate_seconds value="3600" comment="emission interval (s). should be >= dt" />
<rate_trcPerEmission value="10" comment="tracers released per interval" />

<!-- 3) rate - direct emission frequency in Hz (tracers per second) -->
<rate value="0.0333" comment="emission rate (Hz). 0.0333 ≈ 1 tracer / 30 s" />

<!-- 4) rateTimeSeries - time-varying rate from .dat or .csv file -->
<rateTimeSeries>
  <file name="data/discharge_example.dat" />
  <scale value="1.0" comment="optional multiplier on rate values" />
</rateTimeSeries>

<rateTimeSeries>
  <file name="data/discharge_example.csv" comment="columns: time, rate" />
  <scale value="1.0" comment="optional multiplier on rate values" />
</rateTimeSeries>
    
```

### .dat file structure

```

TIME_CYCLE           : 0
TIME_UNITS           : SECONDS
SERIE_INITIAL_DATA   : 2020 01 02 00 00 00

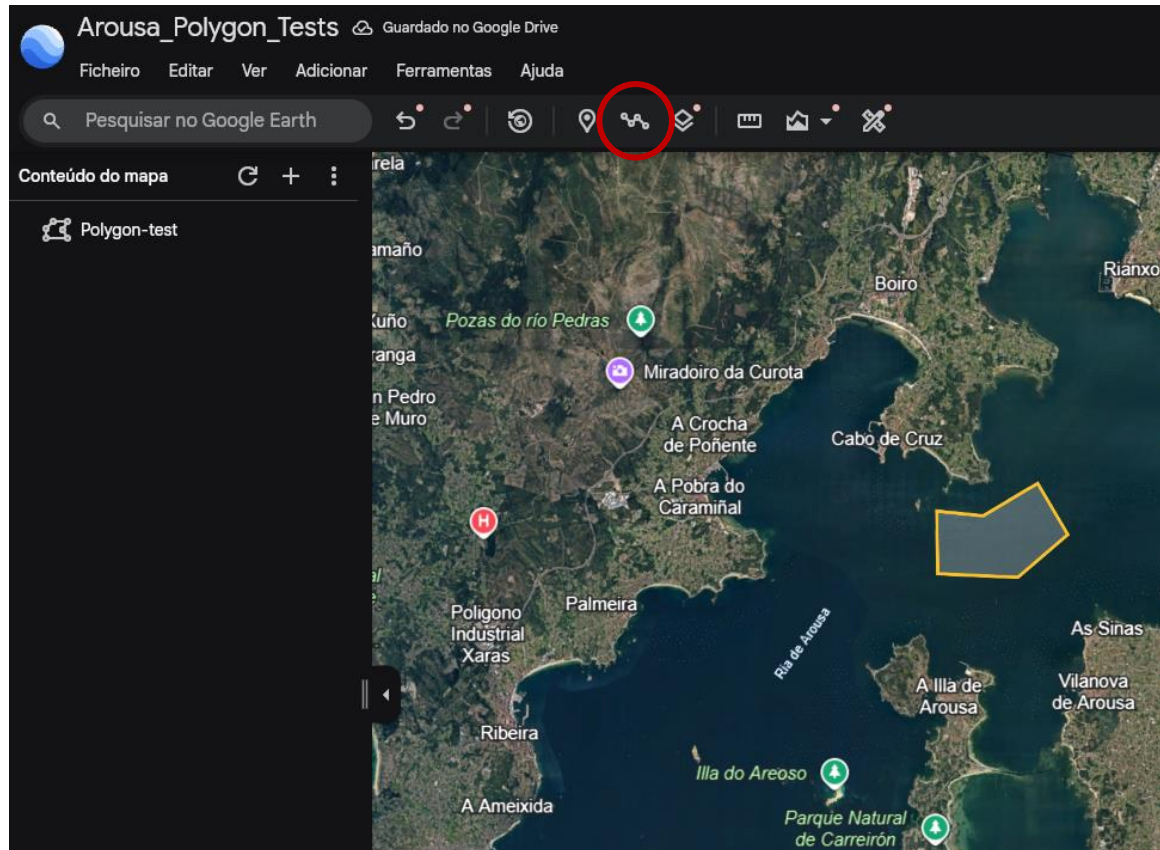
!time                rate
<BeginTimeSeries>
3600                  0.002
7200                  0.005
86400                 0.01
172800                0.006
345600                0.003
691200                0.0
<EndTimeSeries>
    
```

### .csv file structure

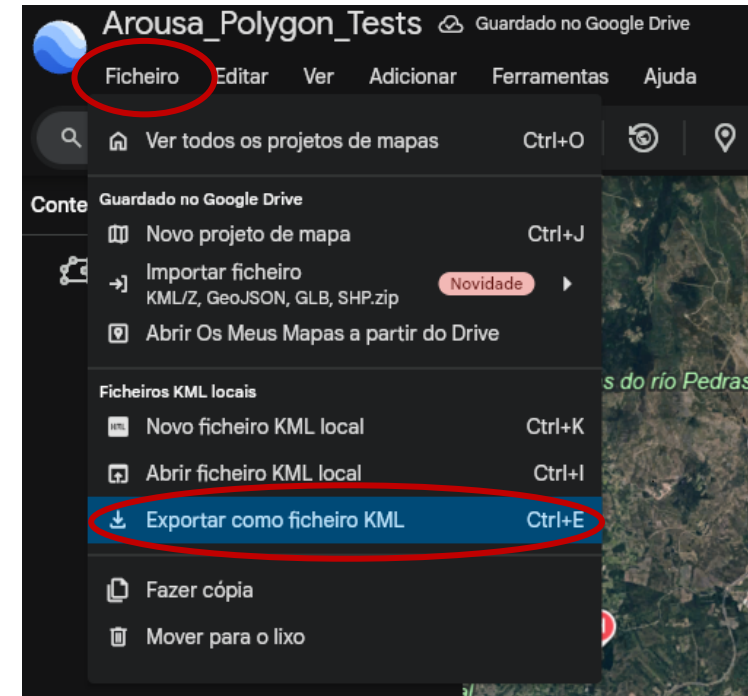
```

time,rate
0,0.001
3600,0.003
86400,0.008
259200,0.004
432000,0.002
691200,0.0
    
```

## > Create your own polygon sources in Google Earth (I)



1 Polygon per file



## > Create your own polygon sources in Google Earth (II)

polygon.kml → Structure

```
<?xml version="1.0" encoding="UTF-8"?>
<kml ...>
  <Document ...>
    ...
    <Placemark>
      ...
      <Polygon>
        <outerBoundaryIs>
          <LinearRing>
            <coordinates>
              ...
            </coordinates>
          </LinearRing>
        </outerBoundaryIs>
      </Polygon>
    </Placemark>
  </Document>
</kml>
```

Change format  
.kml → .kmz

polygon.kmz → Structure

```
<kml>
  <Document>
    <Placemark>
      <Polygon>
        <outerBoundaryIs>
          <LinearRing>
            <coordinates>
              ...
            </coordinates>
          </LinearRing>
        </outerBoundaryIs>
      </Polygon>
    </Placemark>
  </Document>
</kml>
```

# Second simulation!



 ***ParaView***

## > Materials and associated processes

| Process category          | Process ( $p$ )       | Base | Paper | Plastic | Detritus |
|---------------------------|-----------------------|------|-------|---------|----------|
| Default                   | Advection             | ✓    | ✓     | ✓       | ✓        |
|                           | Turbulence            | ✓    | ✓     | ✓       | ✓        |
|                           | Beaching              | ✓    | ✓     | ✓       | ✓        |
| Input data<br>activated   | Windage               | ✓    | ✓     | ✓       |          |
|                           | Stokes drift          | ✓    | ✓     | ✓       | ✓        |
|                           | Biofouling            |      |       | ✓       |          |
| Specifically<br>activated | Buoyancy              |      | ✓     | ✓       | ✓        |
|                           | Resuspension          |      | ✓     | ✓       | ✓        |
|                           | Degradation-linear    |      | ✓     | ✓       |          |
|                           | Degradation-Nonlinear |      |       |         | ✓        |

## > Windage process

- 10m wind forcing →

```
<inputDataDir name="nc_fields/winds" type="meteorology" />
```

- Exponential decay

$$K_i^{\text{wind}}(x_{i,j}, t) = C^{\text{wind}} \cdot v_{i,j}^{10} \cdot \exp(-10 \cdot Z_i^{\text{wind}}), \quad j = 1, 2$$

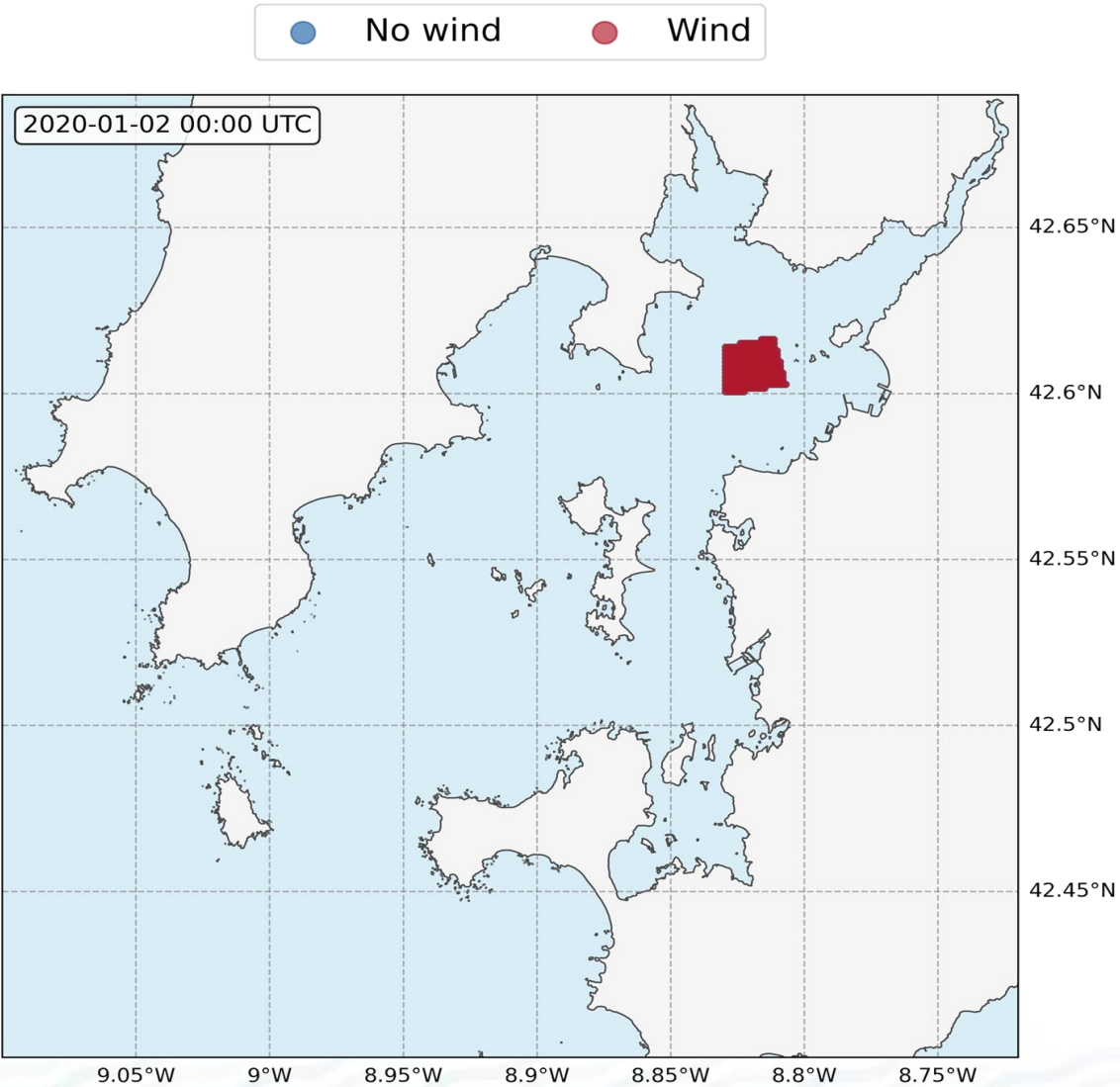
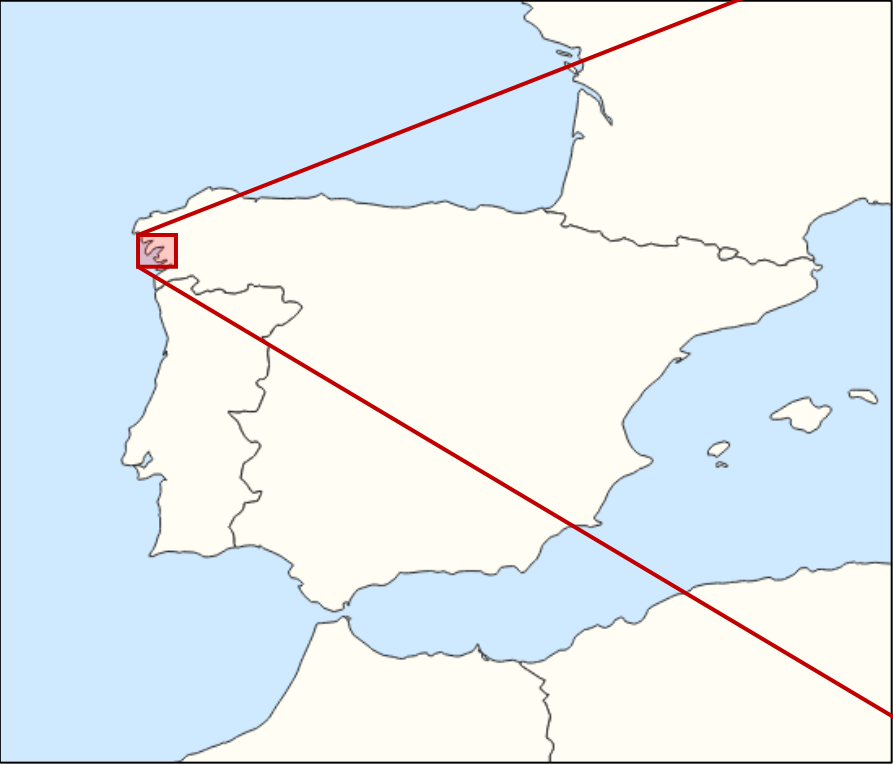
- Mobile coordinate system  
(5 cm surface buffer)

$$Z_i^{\text{wind}} = \max[0, -x_3 + (\text{SeaLevel} - 0.05)]$$

- Wind coefficient ( $C^{\text{wind}}$ ) →

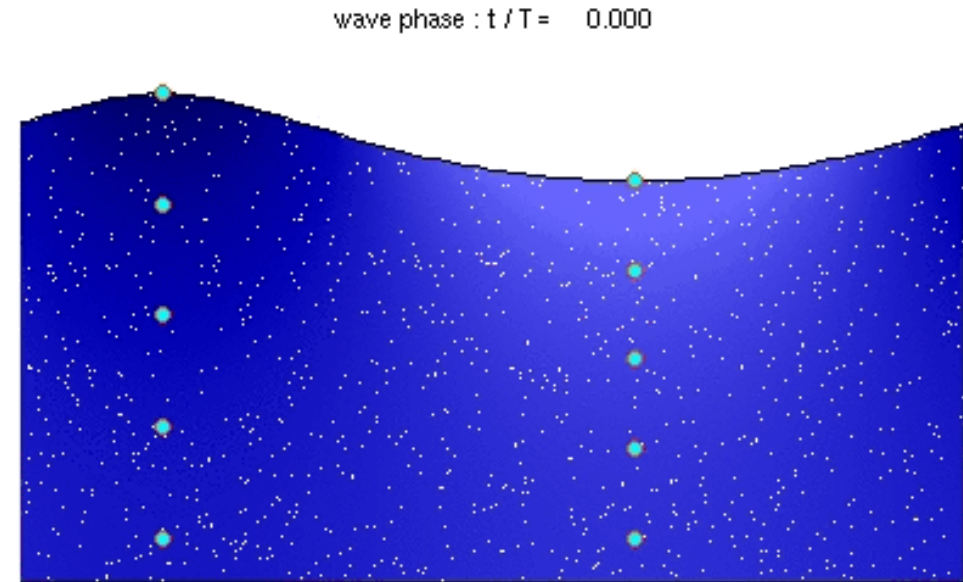
```
<WindDragCoeff value="0.03" comment="Default = 0.0" units_comment="" />
```

> Simulation with/without wind



## > Stokes Drift process (I) – Physical Concept & Inputs

- **Net horizontal transport** at the sea surface in the wave propagation direction.
- Generates a compensating **Eulerian return current** near the coast due to mass conservation.
- **Adaptive data ingestion:** Processes either explicit Stokes velocities or raw wave characteristics.
- Managed via a global **multiplicative calibration coefficient** ( $C^{\text{wave}}$ ) defined in the XML.



Input Variables:

- $H \rightarrow$  Wave Height
- $T \rightarrow$  Wave Period
- $D \rightarrow$  Wave Direction

```
<StokesDriftCoeff value="0.5" comment="Default = 0.0" units_comment="" />
```

## > Stokes Drift process (II) – Direct Velocity Ingestion

```
<inputDataDir name="nc_fields/waves" type="waves"/>
```



Stokes components `stokes_u`, `stokes_v`  
calculated with an external model.

$$\mathcal{K}_i^{\text{stokes}}(x_{i,j}, t) = C^{\text{wave}} \cdot v_{i,j}^{\text{sd}} \cdot \exp(-Z_i^{\text{sd}}), \quad j = 1, 2$$

$$Z_i^{\text{sd}} = \max[0, -x_{i,3}]$$

- Return velocity is assumed to be already resolved by the external wave model.

## > Stokes Drift process (III) – Reconstruction from Wave Characteristics

```
<inputDataDir name="nc_fields/waves" type="waves"/>
```

→ (H,T,D) →

a: amplitude

$\omega$ : angular frequency

n: wave number

- Deep Water Limit ( $H > \lambda/2$ ):

$$v^{sd} = a^2 \cdot \omega \cdot n \cdot \exp(-2nZ)$$

→

Airy Linear Approximation

- Finite Depth ( $H \leq \lambda/2$ ):

$$u_s = \frac{a^2 \omega n \cosh[2n(Z - H_w)]}{2 \sinh^2(nH_w)}$$

$$u_E = \frac{a^2 \omega \sinh(2nH_w)}{4H_w \sinh^2(nH_w)}$$

→

$$\mathbf{v}^{sd} = \mathbf{u}_s - \mathbf{u}_E$$

- Final projection and constraints:

$$\begin{aligned} \mathcal{K}_x &= C^{wave} \cdot v^{sd} \cdot \cos(D) \\ \mathcal{K}_y &= C^{wave} \cdot v^{sd} \cdot \sin(D) \end{aligned}$$

## > Materials and associated processes

| Process category          | Process ( $p$ )       | Base | Paper | Plastic | Detritus |
|---------------------------|-----------------------|------|-------|---------|----------|
| Default                   | Advection             | ✓    | ✓     | ✓       | ✓        |
|                           | Turbulence            | ✓    | ✓     | ✓       | ✓        |
|                           | Beaching              | ✓    | ✓     | ✓       | ✓        |
| Input data<br>activated   | Windage               | ✓    | ✓     | ✓       |          |
|                           | Stokes drift          | ✓    | ✓     | ✓       | ✓        |
|                           | Biofouling            |      |       | ✓       |          |
| Specifically<br>activated | Buoyancy              |      | ✓     | ✓       | ✓        |
|                           | Resuspension          |      | ✓     | ✓       | ✓        |
|                           | Degradation-linear    |      | ✓     | ✓       |          |
|                           | Degradation-Nonlinear |      |       |         | ✓        |



## materialTypes.xml

### Structure

```
<materials>
  <plastic>
    <plastic_type_1>
      ...
    <plastic_type_1>
    <plastic_type_2>
      ...
    <plastic_type_2>
  </plastic>
  ...
  <detritus>
    ...
  </detritus>
  ...
</materials>
```

### Example of plastic material type

```
<pellet_ldpe>
  <particulate value="1" />
  <density value="920.0" />
  <radius value="0.01" />
  <volume value="4.18879E-06" />
  <area value="1.256637E-03" />
  <property key="condition" value="1.0" />
  <property key="degradation_rate" value="0.0" />
</pellet_ldpe>
```

## > Buoyancy process (I) — Temperature and salinity fields.

- With Temperature and Salinity fields → `<inputDataDir name="nc_fields/waterProperties" type="waterProperties"/>`

$$(T, S \rightarrow \rho_{SW}, \nu_{SW})$$

$$\rho_{SW}(S, T) = \rho_W(T) + A(T) \cdot S + B(T) \cdot S^{3/2} + C(T) \cdot S^2$$

$$\nu_{SW}(S, T) = \frac{\mu_W(T) \cdot (1 + A_v \cdot S + B_v \cdot S^2)}{\rho_{SW}}$$

- Without Temperature and salinity → Fallback values: 
$$\left\{ \begin{array}{l} \rho_{SW} = 1025.0 \text{ kg/m}^3 \\ \nu_{SW} = 1.0 \times 10^{-6} \text{ m}^2/\text{s} \end{array} \right.$$

## > Buoyancy process (II) — The Physics Kernel

- Density and radius from *materialTypes.xml*



- Dimensionless diameter:

$$d_i^* = d_i \cdot \left( \frac{|\rho_i - \rho_{SW}|}{\rho_{SW}} \cdot \frac{g}{v_{SW}^2} \right)^{1/3}$$

- Particle velocity:

$$w_S = \frac{v_{SW}}{d_i} \cdot d_i^{*3} \cdot \left( 38.1 + 0.93 \cdot d_i^{*12/7} \right)^{-7/8}$$

- Sign evaluation by comparison between  $\rho_i$  and  $\rho_{SW}$   $\rightarrow$   $\mathcal{K}_i^{\text{buoy}}(x_3, t) = \text{sign}Z_i \cdot w_S$

# Third simulation!



 ***ParaView***



# Activity 5: Complementary processes and tracer types

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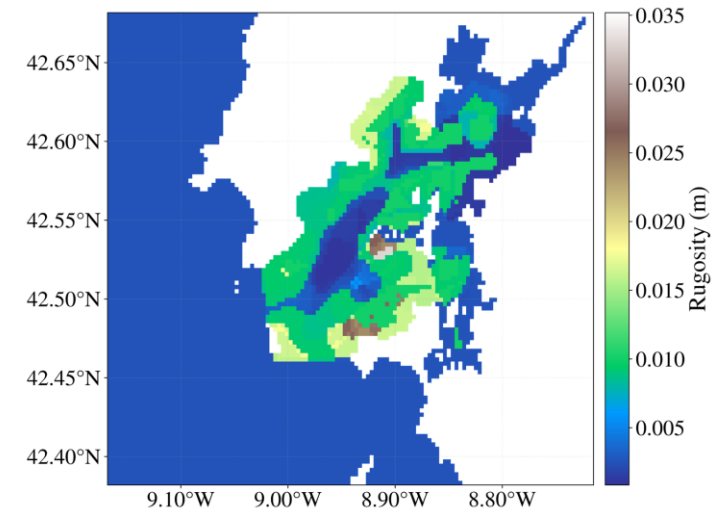
## > Resuspension (I) — Variable definitions

- Boundary Layer (bl) definition → `<BedLoadThickness value="0.04" comment="Default = 0.04" units_comment="m" />`
- Rugosity definition ( $\varepsilon_0$ ) → `<RugosityinNetcdf value="0" comment=" 1:true, 0:false (Default = 0)"/>`

Value = 0: Rugosity defined in xml

```
<Rugosity value="0.0025"  
  comment=" Default = 0.0025"  
  units_comment="m"  
>
```

Value = 1: Rugosity map in netCDF



- Median diameter (D50) → `<D50inNetcdf value="0" comment="1:true, 0:false. (Default=0)"/>`

## > Resuspension (II) — Velocity reduction and sedimentation

- Velocity reduction:

$$\mathcal{K}_{k,i}^{\text{resusp}}(x_j, t) = \eta(\tau_{\text{bed}}) \cdot v_{i,j}^{\text{bl}}(\mathbf{x}, t), \quad j = 1, 2, 3$$

○  $\tau_{\text{bed}}$  calculation:  $v_{i,j} = \frac{v_{i,j}^*}{\kappa} \ln \left( \frac{x_{i,3} - H_i^b}{\varepsilon_0} \right), \quad j = 1, 2, 3$

○ Promediated Bedload velocity:  $v_{i,j}^{\text{bl}} = \frac{v^*}{\kappa} \left[ \ln \left( \frac{bl}{\varepsilon_0} \right) - 1 - \frac{\varepsilon_0}{bl} \right]$

$\tau_{\text{bed}} = \rho \cdot (v^*)^2$

$v^*$

- Sedimentation component:

$$\mathcal{P}_i(t) = \mathcal{P}_{0,\text{rsp}} \left[ \exp \left( - \frac{\text{age}_i}{\tau_{\text{rsp}}} \right) \right]$$

```
<ResuspensionProb value="100.0" comment="Default = 0%" units_comment="%" />
<ResuspensionResidenceTime value="8640000.0" comment="Default = 10*24*60*60" units_comment="sec" />
```

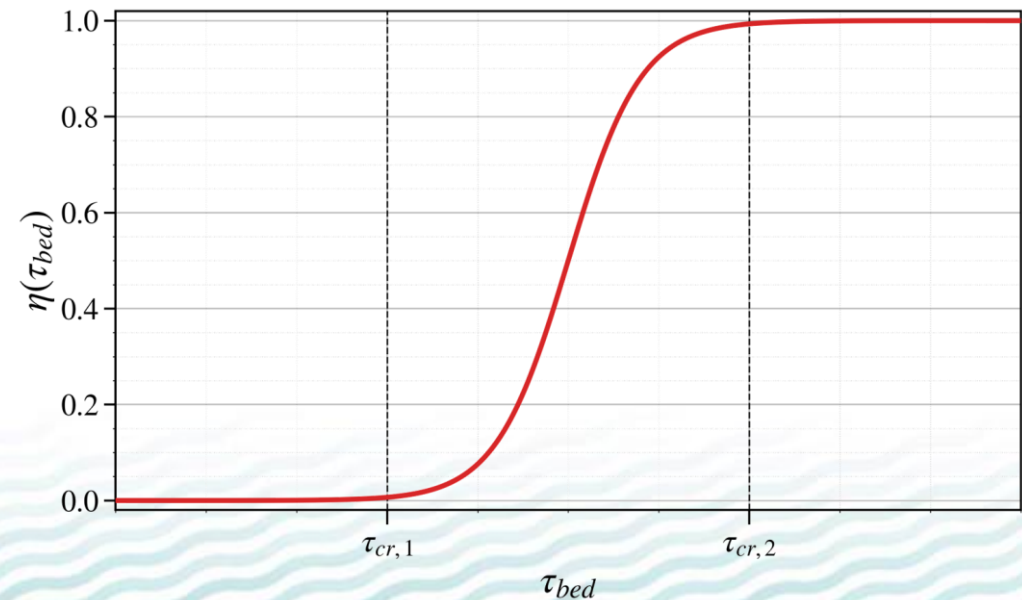
## > Resuspension (III) — Resuspension criteria

- Different motion regimes:

$$\eta(\tau_{bed}) = \begin{cases} 0, & \tau_{bed} \leq \tau_{cr,1} \\ \alpha(\tau_{bed}), & \tau_{cr,1} < \tau_{bed} < \tau_{cr,2} \\ 1, & \tau_{bed} \geq \tau_{cr,2} \end{cases} \begin{array}{l} \rightarrow \text{No motion} \\ \rightarrow \text{Bedload transport} \\ \rightarrow \text{Full resuspension} \end{array}$$

- Unified function:

$$\eta(\tau_{bed}) = \left[ 1 + \exp \left( -\gamma \frac{\tau_{bed} - \frac{\tau_{cr,2} + \tau_{cr,1}}{2}}{\tau_{cr,2} - \tau_{cr,1}} \right) \right]^{-1}$$



## > Resuspension (IV) — Shear stress options

$$\tau_{cr,j} = \theta_{cr,j} \cdot (\rho_i - \rho_{sw}) \cdot g \cdot d_i, \quad j = 1,2$$

```
<ResuspensionCriticalShearMethod value="2" comment="Default = 1" />
```

- Value = 1 → Constant values defined in *xml*:

```
<ResuspensionCriticalShear1 value="0.01" comment="Default = 0.10" units_comment="N/m2" />  
<ResuspensionCriticalShear2 value="0.045" comment="Default = 0.45" units_comment="N/m2" />
```

- Value = 2 → Shear Stress dependent on diameter:

$$\theta_{cr,1} = \frac{0.3}{1 + 1.2 d_i^*} + 0.055[1 - \exp(-0.02 d_i^*)]$$
$$\theta_{cr,2} = \frac{0.3}{1 + d_i^*} + 0.10[1 - \exp(-0.05 d_i^*)]$$

- Value = 3 → Method 2 + Hiding exposure

## > Degradation in organic matter

$$\mathcal{D}(t) = (1 - \beta \cdot t) \cdot [\alpha e^{-\mu_1 t} + (1 - \alpha) e^{-\mu_2 t}]$$

- $\beta \rightarrow$  `<Lin0_DegradationRate value="0.0" comment="Default = 0.030/86400" units_comment="Input(1/day)" />`
- $\alpha \rightarrow$  `<Exp0_DegradationRate value="0.14" comment="Default = 0.14" units_comment=" unit=1" />`
- $\mu_1 \rightarrow$  `<Exp1_DegradationRate value="0.514" comment="Default = 0.514/86400" units_comment="Input(1/day)" />`
- $\mu_2 \rightarrow$  `<Exp2_DegradationRate value="0.40" comment="Default = 0.040/86400" units_comment="Input(1/day)" />`

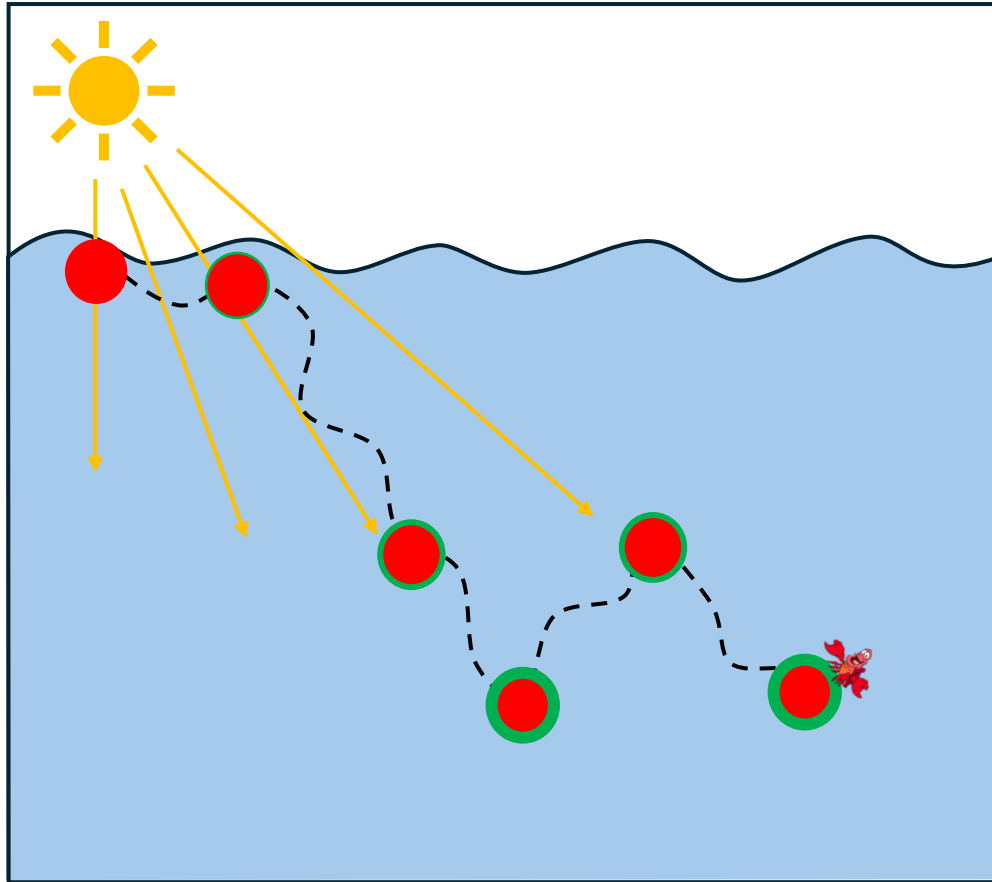
$$\frac{dV_i}{dt} = \frac{1}{\Delta t} \left( \frac{\mathcal{M}_i(0), \mathcal{D}_i(t + \Delta t) \cdot \mathcal{L}(T) - \mathcal{M}_i(t)}{\rho_i(t)} \right) \longrightarrow \mathcal{L}(T): \text{Temperature correction}$$

# Detritus simulation!



 ***ParaView***

## > Biofouling (I) — Physical Concept & Inputs



Algal and bacterial growth around debris surfaces, mainly depending on:

- Temperature and salinity
- Bacterial and algal concentration
- Light intensity
- Grazing mortality
- Algal respiration



## > Biofouling (II) — Theoretical discussion

Simple spherical particles + biofilm grows uniformly around the plastic surface:

$$\rho_{tot}(z, t) = \frac{r_{pl}^3 \rho_{pl} + [(r_{pl} + t_{bf})^3 - r_{pl}^3] \rho_{bf}}{(r_{pl} + t_{bf})^3}$$

$$t_{bf} = \left( \frac{3V_{tot}}{4\pi} \right)^{1/3} - R_0, \quad V_{tot} = V_b + V_0, \quad V_0 = \frac{4\pi R_0^3}{3}, \quad V_b = V_A S_0 A$$

Algal growth,  $\frac{dA}{dt}$ , was dynamically modelled as:

$$\frac{dA}{dt} = \underbrace{\frac{\beta_A A_A}{S}}_{\text{Collision}} + \underbrace{\mu(T, I) A}_{\text{Growth}} - \underbrace{m_A A}_{\text{Mortality}} - \underbrace{Q_{10}^{\frac{T-20}{10}} R_A A}_{\text{Respiration}}$$

# > Biofouling (III) — Simulation behaviour

